



MARIO C. CALATAN

Utility Systems Engineer | GIS & SCADA Specialist | Smart Grid & Web GIS Developer

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PROFESSIONAL SUMMARY

Electrical Engineer with over 10 years of experience in electric distribution utilities, specializing in Smart Grid modernization, SCADA/GIS integration, Advanced Metering Infrastructure (AMI), and utility digital transformation initiatives. Experienced in leading cross-functional engineering teams, system modernization programs, and development of enterprise-grade utility applications that improve reliability, operational efficiency, compliance, and decision-making.

Strong hybrid background combining power systems engineering, utility operations, software development, databases, and AI-assisted technologies to deliver end-to-end solutions supporting future-ready electric utility operations.

CORE COMPETENCIES

Utility Engineering & Smart Grid

- Electric Distribution Planning & Design (up to 23kV)
- Smart Grid Technologies (SCADA, OMS, DMS, ADMS, AMI, FLISR)
- Distribution Automation and Utility Modernization
- Meter Data Management Systems (MDMS)
- Distribution System Reliability and Performance Analysis
- Utility Asset Planning and Infrastructure Development

Power Systems Engineering

- Load Flow Analysis
- Short Circuit Studies
- Arc Flash Analysis
- Protection Coordination Studies
- Feeder Capacity Assessment
- Fault Investigation and System Analysis
- Transformer Loading and Capacity Assessment
- Voltage Profile and Feeder Optimization

Relay & Protection Systems

- Protection relay coordination support
- Hands-on experience with ABB, GE, SEL, G&W, and Eaton equipment
- Recloser and feeder protection applications
- Fault isolation and switching strategies

Software Development & Automation

- Web-based Utility Systems Development
- Full-stack Development (PHP, Python, JavaScript, React, Node.js)
- Utility Data Integration and Analytics
- Database Design and Optimization
- AI-Assisted Development and Automation

Databases

- PostgreSQL
- MySQL/MariaDB

- Oracle
- Microsoft SQL Server

Communication Protocols

- DNP3
- Modbus
- SCADA Data Communications

AI & Emerging Technologies

- GitHub Copilot
- Claude AI
- GPT/OpenAI
- Base44 AI-assisted development tools

SOFTWARE & TECHNICAL TOOLS

Smart Grid Platforms

- Survalent
- QEI
- SEL Technologies

Planning & Engineering Tools

- Synergi Electric
- ETAP
- PVSyst

GIS & Design Tools

- QGIS
- PostGIS
- ArcGIS
- AutoCAD
- SketchUp
- Canva

Productivity Platforms

- Microsoft Office Suite
(Word, Excel, PowerPoint)

KEY SYSTEMS & PLATFORMS DEVELOPED

Led the design and development of multiple in-house enterprise applications supporting utility operations and digital transformation initiatives:

- Network Management System (NMS)
- Web GIS Platform with modules including:
 - Vegetation Management
 - Damage Assessment
 - Thermal Scan Inspection Management
- Meter Installation Management System
- Meter Replacement Management System
- Technical Repository System
- Compliance Reporting System
- Transformer Management System

- Meter Data Management System (MDMS)
 - Calibration Management System
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PROFESSIONAL EXPERIENCE

System Planning & Design Engineer

BENECO | 2024 – Present

- Lead planning and design activities for electric distribution network development and infrastructure expansion initiatives
 - Manage engineering teams responsible for CAPEX/OPEX planning, inspections, forecasting, and project prioritization
 - Conduct power system studies including:
 - Load Flow Analysis
 - Short Circuit Studies
 - Protection Coordination Studies
 - Reliability Analysis
 - Lead As-Planned and As-Built engineering activities
 - Develop technical reports and management presentations supporting strategic decisions
 - Serve as Chairman of the Technical Working Group under the Bids and Awards Committee (BAC) responsible for:
 - Technical specification review
 - Bid evaluation
 - Recommendation preparation
 - Support long-term infrastructure planning and utility modernization initiatives
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Special Equipment & Metering Officer

BENECO | 2020 – 2024

- Led implementation initiatives for Advanced Metering Infrastructure (AMI) including Proof-of-Concept, pilot, and deployment phases
 - Developed and implemented multiple in-house utility systems supporting metering and operational functions
 - Led deployment of Meter Data Management System (MDMS) solutions
 - Performed testing and technical evaluation of:
 - Meters
 - Transformers
 - CT/PT systems
 - Line equipment
 - Supported testing and commissioning activities
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Construction & Maintenance Officer

BENECO | 2019 – 2020

- Managed 24/7 field operations and maintenance activities
- Led emergency response coordination and service restoration efforts

- Supervised field crews and coordinated operational priorities
 - Provided technical guidance for construction and maintenance activities
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SCADA & GIS Development Engineer

BENECO | 2013 – 2019

- Modernized SCADA infrastructure supporting OMS/DMS readiness initiatives
 - Led GIS enhancement projects and utility network modeling improvements
 - Integrated operational systems to improve situational awareness and data visibility
 - Supported Smart Grid and utility digital transformation initiatives
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GIS Data Engineer

BENECO | 2011 – 2013

- Conducted GIS data gathering and validation activities
 - Supported GIS implementation and utility asset integration projects
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ADMINISTRATIVE & LEADERSHIP EXPERIENCE

- CAPEX and OPEX preparation, justification, and monitoring
 - Team leadership and workforce planning
 - Stakeholder coordination across operations, vendors, regulators, and communities
 - Procurement support and resource planning
 - Technical report preparation and executive presentations
 - Project documentation and compliance reporting
 - Policy implementation and ISO/IMS process alignment
 - BAC Technical Working Group leadership
 - Mentoring and cross-functional coordination
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EDUCATION

Master in Urban Management (*Ongoing*)
Benguet State University | 2020 – Present

Bachelor of Science in Electrical Engineering
Saint Louis University | 2006 – 2011

ADDITIONAL INFORMATION

- Willing to relocate internationally with family subject to employer-supported arrangements
- Strong background in Smart Grid and utility digital transformation initiatives
- Combines engineering leadership with software and analytics expertise
- Open to utility engineering, planning, smart grid, and digital transformation opportunities